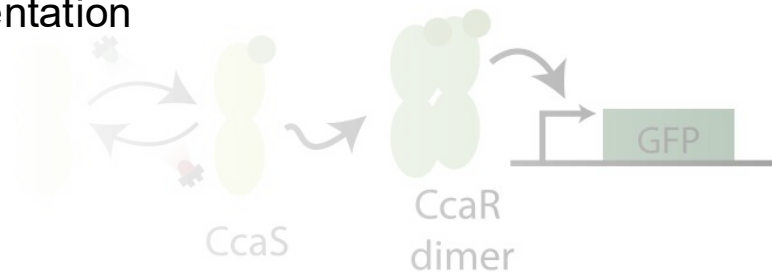


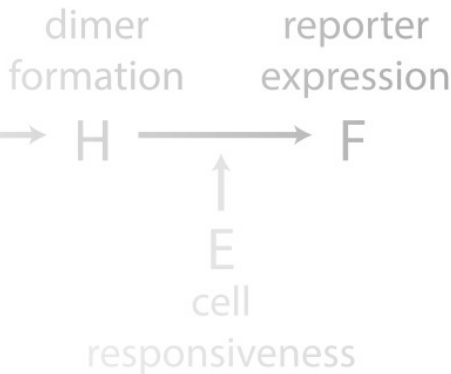
Gene Expression Predict and Control

Tianrui Qi
Apr 16, 2026

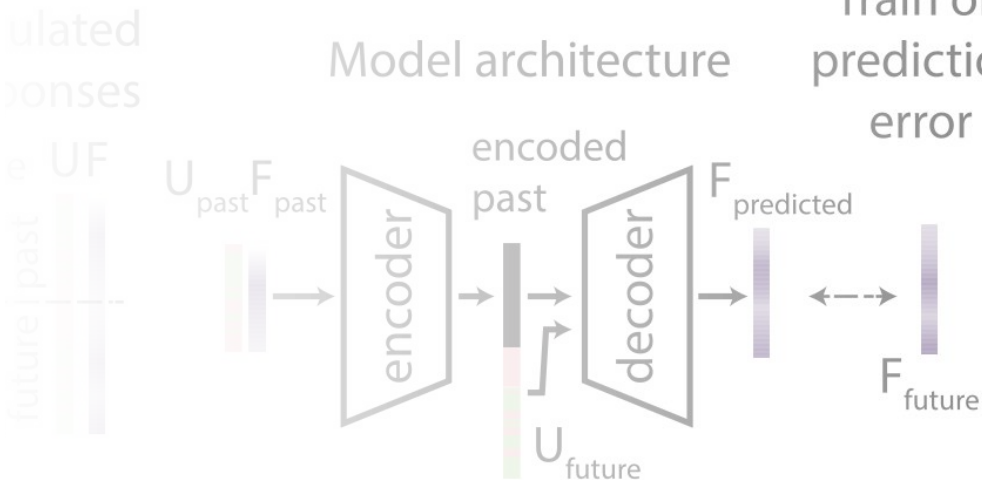
Example inducible system: CcaSR



Circuit architecture

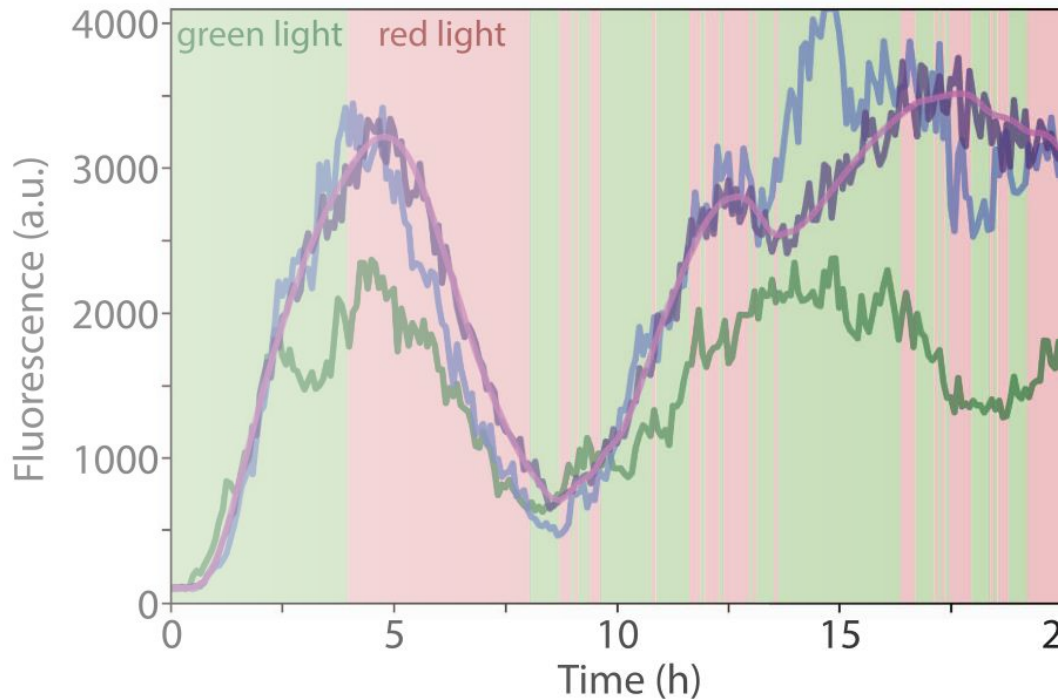


Model architecture



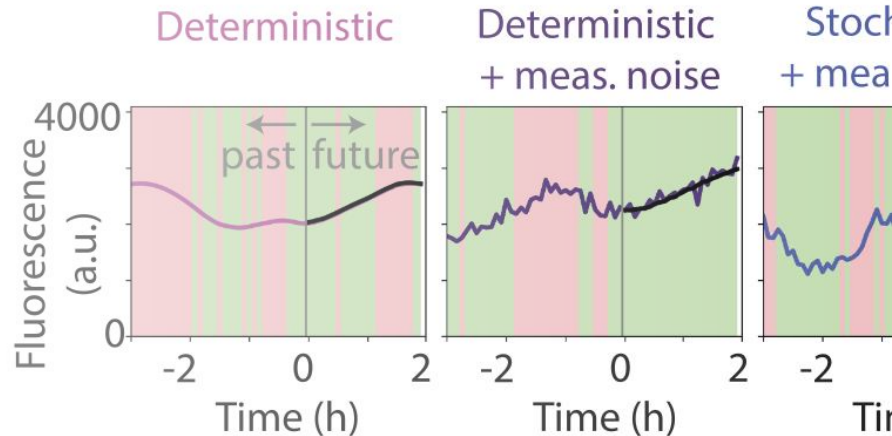
C

Sample simulations



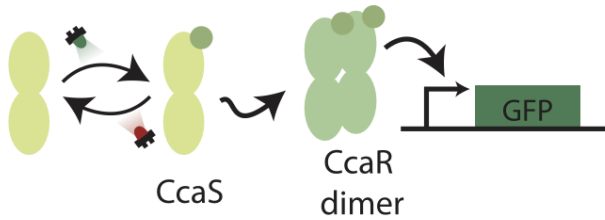
E

Median prediction ex



Background

A Example inducible system: CcaSR



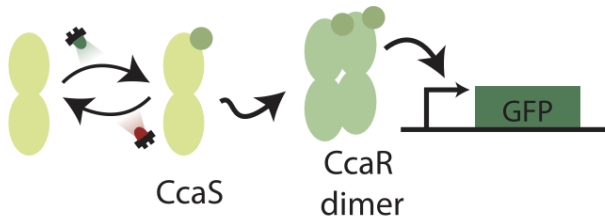
Optogenetic System: CcaS/CcaR

- Green light → activates CcaS (phosphorylation) → CcaR forms dimers → dimers bind promoter → GFP expression ↑
- Red light → deactivates CcaS (dephosphorylation) → dimers disassemble → GFP diluted out ↓

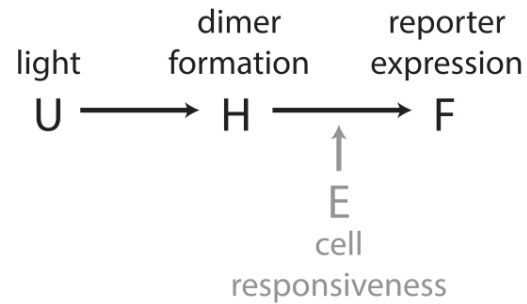
Around 12 min delay between light stimulus and observable response, due to (de)phosphorylation kinetics and dimer (dis)assembly dynamics.

Background

A Example inducible system: CcaSR



B Circuit architecture



1. Nonlinear propagation ($H \rightarrow F$)

Sigmoidal saturating relationship between an activator (H) and its output (F), described by Hill function.

The Hill function amplifies H fluctuations/noise in a state-dependent way.

2. Cell responsiveness (E modulates $H \rightarrow F$ gain)

E scales F production rate: high/low E $\rightarrow$ faster/slower F production for same H.

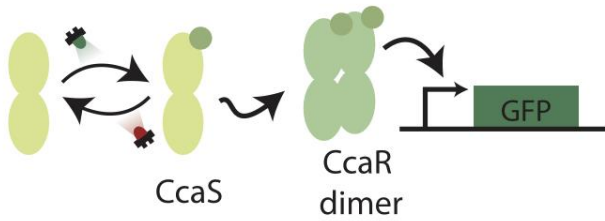
Since E is unobservable and drifts stochastically (Poisson distribution), the effective $H \rightarrow F$ gain is constantly shifting.

3. Intrinsic noise ($U \rightarrow H, H \rightarrow F, E$)

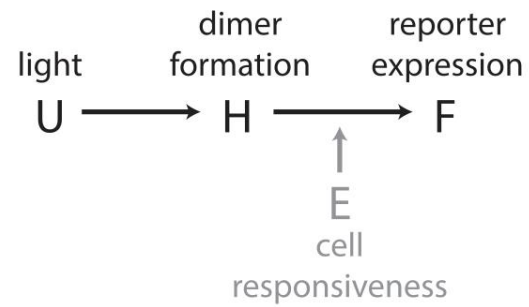
Stochastic gene expression is a discrete random process (exponential distribution). Identical light inputs produce different H trajectories every time.

Background

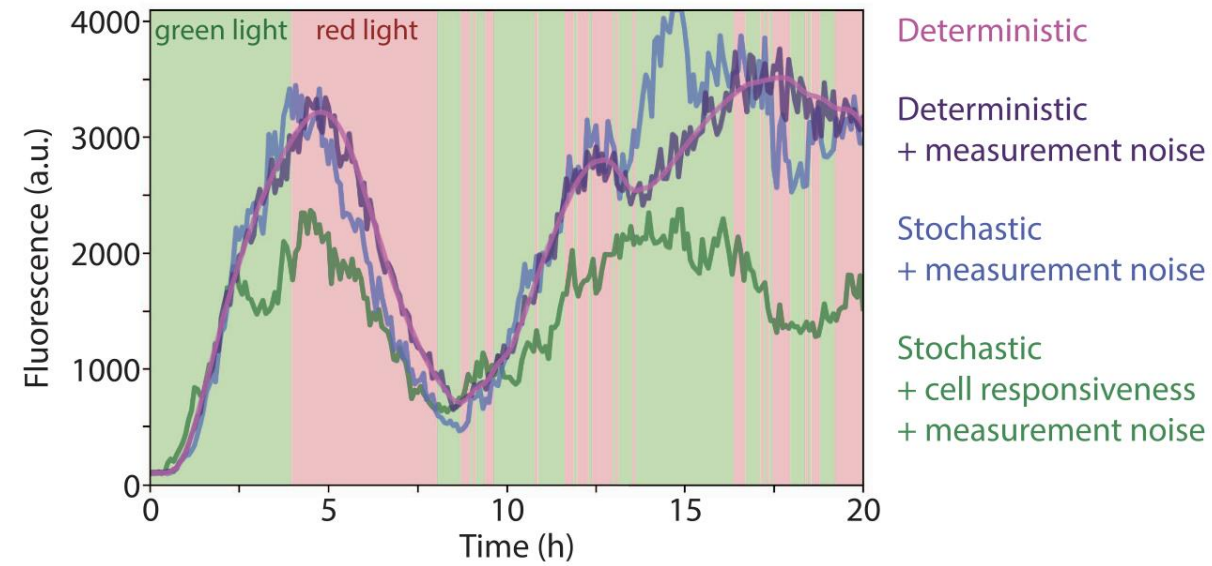
A Example inducible system: CcaSR



B Circuit architecture

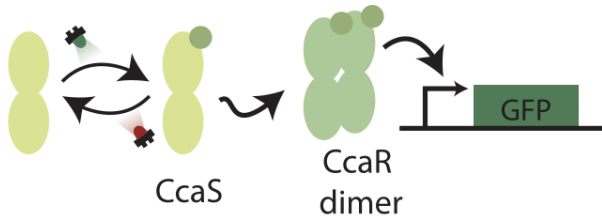


C Sample simulations

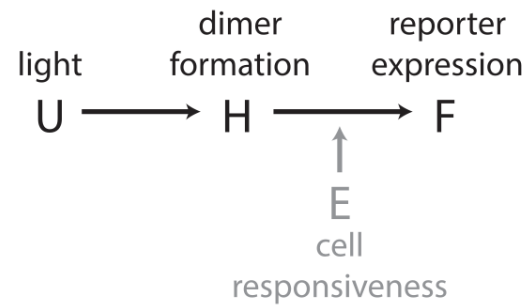


Background

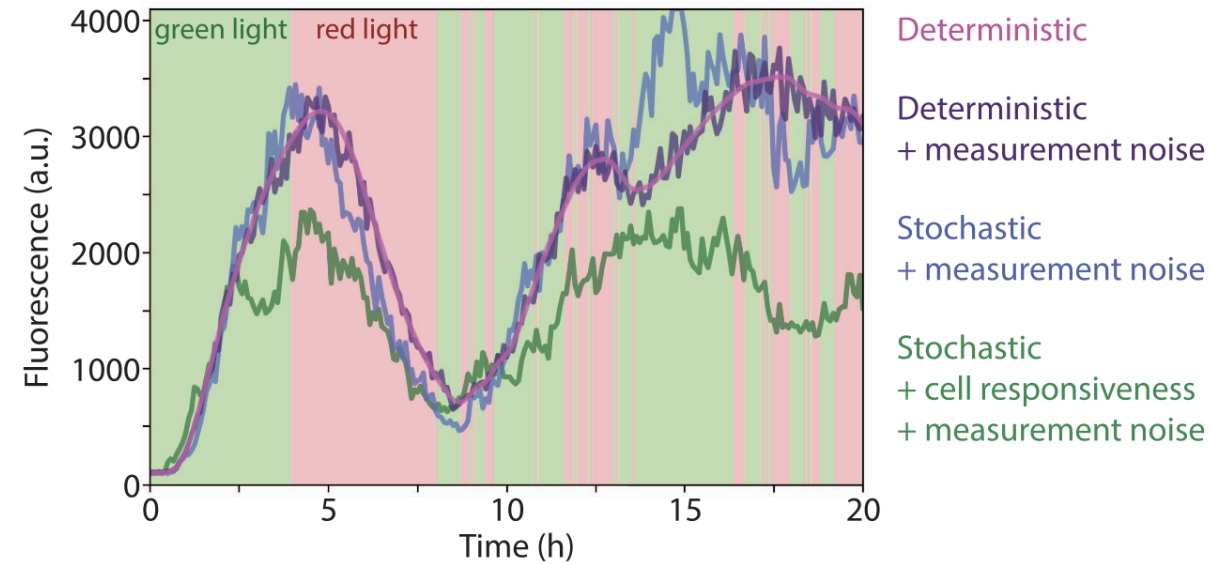
A Example inducible system: CcaSR



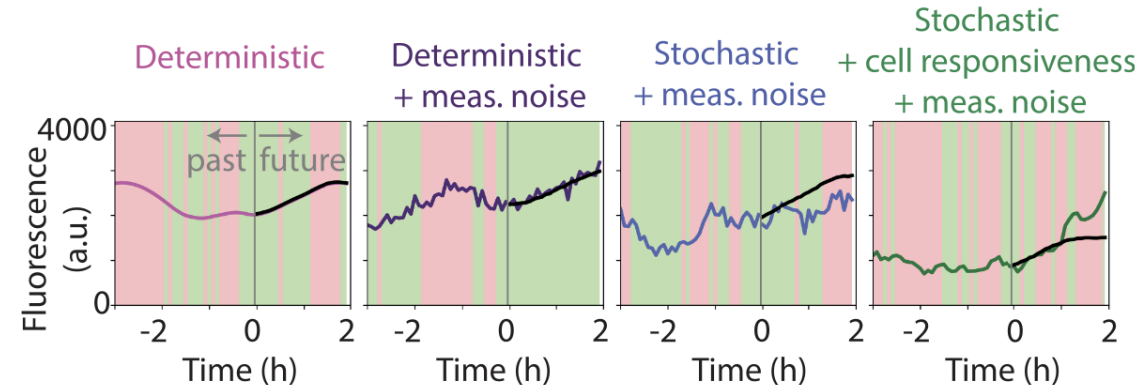
B Circuit architecture



C Sample simulations

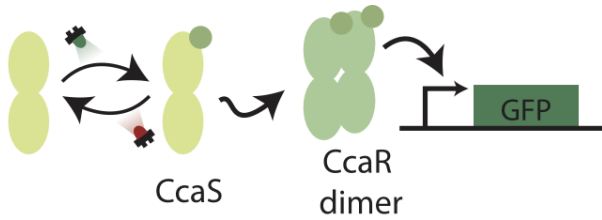


E Median prediction examples

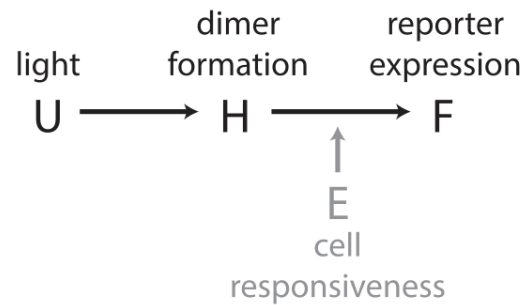


Background

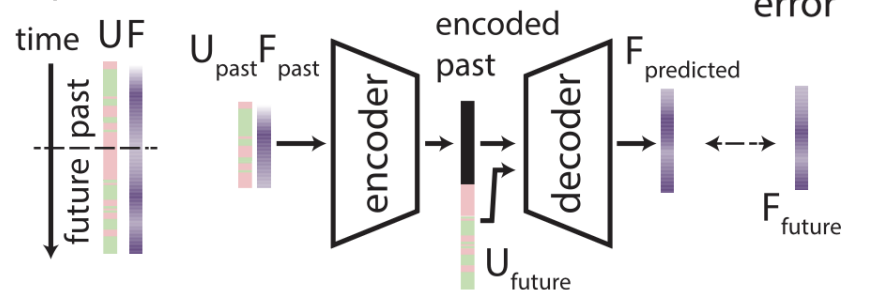
A Example inducible system: CcaSR



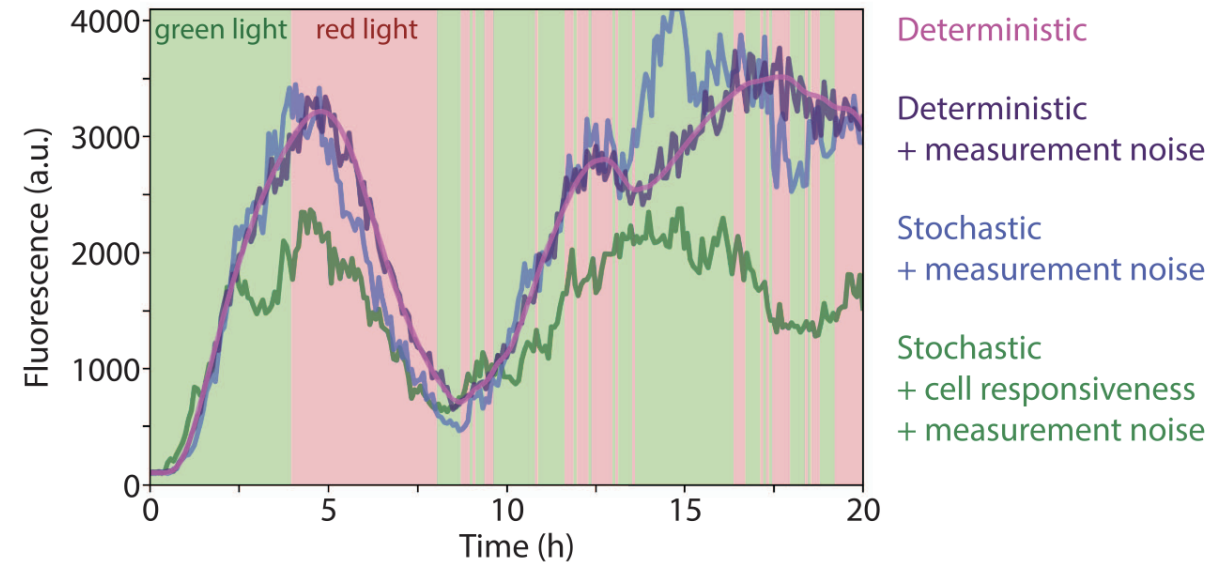
B Circuit architecture



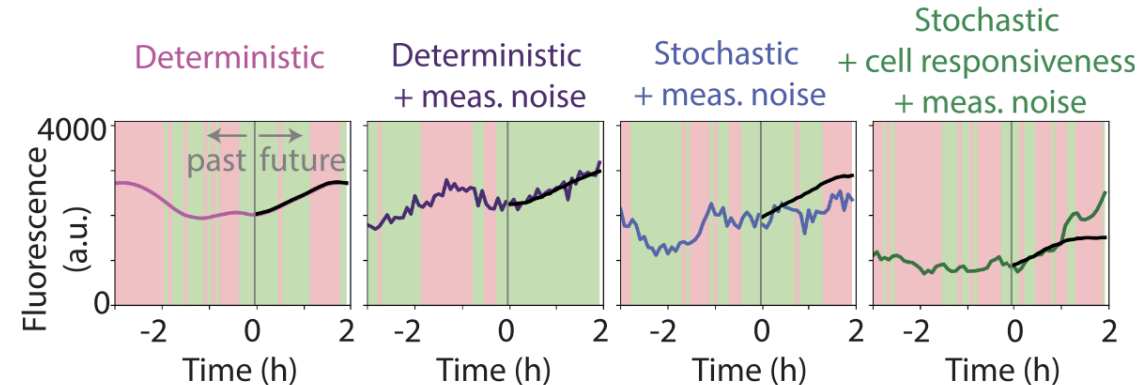
D Simulated responses
Model architecture



C Sample simulations



E Median prediction examples



Background

Previous publications on deep cell control from Dunlop Lab



Deep Neural Networks for Predicting Single-Cell Responses and Probability Landscapes

Published as part of the ACS Synthetic Biology virtual special issue “AI for Synthetic Biology”.

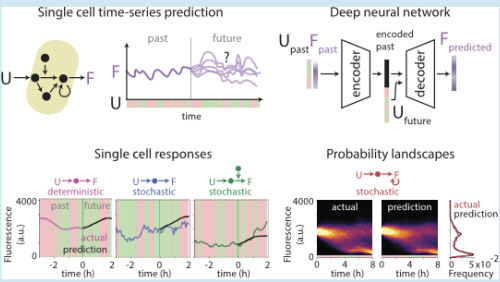
Heidi E. Klumpe,^{||} Jean-Baptiste Lugagne,^{*,||} Ahmad S. Khalil, and Mary J. Dunlop^{*}

Cite This: ACS Synth. Biol. 2023, 12, 2367–2381

Read Online

ABSTRACT: Engineering biology relies on the accurate prediction of cell responses. However, making these predictions is challenging for a variety of reasons, including the stochasticity of biochemical reactions, variability between cells, and incomplete information about underlying biological processes. Machine learning methods, which can model diverse input–output relationships without requiring *a priori* mechanistic knowledge, are an ideal tool for this task. For example, such approaches can be used to predict gene expression dynamics given time-series data of past expression history. To explore this application, we computationally simulated single-cell responses, incorporating different sources of noise and alternative genetic circuit designs. We showed that deep neural networks trained on these simulated data were able to correctly infer the underlying dynamics of a cell response even in the presence of measurement noise and stochasticity in the biochemical reactions. The training set size and the amount of past data provided as inputs both affected prediction quality, with cascaded genetic circuits that introduce delays requiring more past data. We also tested prediction performance on a bistable auto-activation circuit, finding that our initial method for predicting a single trajectory was fundamentally ill-suited for multimodal dynamics. To address this, we updated the network architecture to predict the entire distribution of future states, showing it could accurately predict bimodal expression distributions. Overall, these methods can be readily applied to the diverse prediction tasks necessary to predict and control a variety of biological circuits, a key aspect of many synthetic biology applications.

KEYWORDS: time-series prediction, deep learning models, noise, single cell, gene expression dynamics, bistability



Klumpe et al. ACS Synth. Biol. 12, 2367–2381 (2023)

Background

Previous publications on deep cell control from Dunlop Lab

ACS
SyntheticBiology

pubs.acs.org/synthbio

Research Article

Deep Neural Networks for Predicting Single-Cell Responses and Probability Landscapes

Published as part of the ACS Synthetic Biology *virtual special issue* "AI for Synthetic Biology".

Heidi E. Klumpe,^{||} Jean-Baptiste Lugagne,^{*,||} Ahmad S. Khalil, and Mary J. Dunlop^{*}

 Cite This: *ACS Synth. Biol.* 2023, 12, 2367–2381

 Read Online

ACCESS |

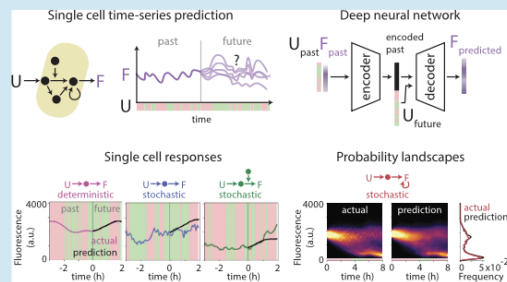
 Metrics & More

 Article Recommendations

 Supporting Information

ABSTRACT: Engineering biology relies on the accurate prediction of cell responses. However, making these predictions is challenging for a variety of reasons, including the stochasticity of biochemical reactions, variability between cells, and incomplete information about underlying biological processes. Machine learning methods, which can model diverse input–output relationships without requiring *a priori* mechanistic knowledge, are an ideal tool for this task. For example, such approaches can be used to predict gene expression dynamics given time-series data of past expression history. To explore this application, we computationally simulated single-cell responses, incorporating different sources of noise and alternative genetic circuit designs. We showed that deep neural networks trained on these simulated data were able to correctly infer the underlying dynamics of a cell response even in the presence of measurement noise and stochasticity in the biochemical reactions. The training set size and the amount of past data provided as inputs both affected prediction quality, with cascaded genetic circuits that introduce delays requiring more past data. We also tested prediction performance on a bistable auto-activation circuit, finding that our initial method for predicting a single trajectory was fundamentally ill-suited for multimodal dynamics. To address this, we updated the network architecture to predict the entire distribution of future states, showing it could accurately predict bimodal expression distributions. Overall, these methods can be readily applied to the diverse prediction tasks necessary to predict and control a variety of biological circuits, a key aspect of many synthetic biology applications.

KEYWORDS: time-series prediction, deep learning models, noise, single cell, gene expression dynamics, bistability



Achieved

- Used LSTM-MLP neural networks to predict future single-cell gene expression trajectories
- Purely computational study, no real experimental data
- Systematically tested factors affecting prediction quality: training set size, amount of past data, circuit complexity (cascade)

Klumpe et al. *ACS Synth. Biol.* 12, 2367–2381 (2023)

Background

Previous publications on deep cell control from Dunlop Lab

ACS
SyntheticBiology

pubs.acs.org/synthbio

Research Article

Deep Neural Networks for Predicting Single-Cell Responses and Probability Landscapes

Published as part of the ACS Synthetic Biology virtual special issue "AI for Synthetic Biology".

Heidi E. Klumpe,^{||} Jean-Baptiste Lugagne,^{*,||} Ahmad S. Khalil, and Mary J. Dunlop^{*}

Cite This: ACS Synth. Biol. 2023, 12, 2367–2381

Read Online

ACCESS |

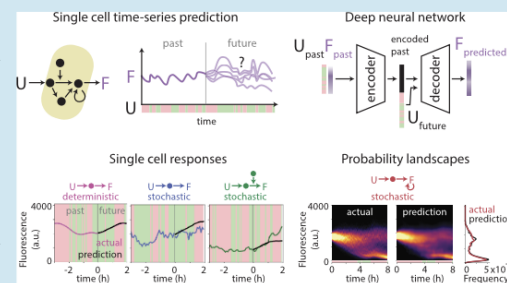
Metrics & More

Article Recommendations

Supporting Information

ABSTRACT: Engineering biology relies on the accurate prediction of cell responses. However, making these predictions is challenging for a variety of reasons, including the stochasticity of biochemical reactions, variability between cells, and incomplete information about underlying biological processes. Machine learning methods, which can model diverse input–output relationships without requiring *a priori* mechanistic knowledge, are an ideal tool for this task. For example, such approaches can be used to predict gene expression dynamics given time-series data of past expression history. To explore this application, we computationally simulated single-cell responses, incorporating different sources of noise and alternative genetic circuit designs. We showed that deep neural networks trained on these simulated data were able to correctly infer the underlying dynamics of a cell response even in the presence of measurement noise and stochasticity in the biochemical reactions. The training set size and the amount of past data provided as inputs both affected prediction quality, with cascaded genetic circuits that introduce delays requiring more past data. We also tested prediction performance on a bistable auto-activation circuit, finding that our initial method for predicting a single trajectory was fundamentally ill-suited for multimodal dynamics. To address this, we updated the network architecture to predict the entire distribution of future states, showing it could accurately predict bimodal expression distributions. Overall, these methods can be readily applied to the diverse prediction tasks necessary to predict and control a variety of biological circuits, a key aspect of many synthetic biology applications.

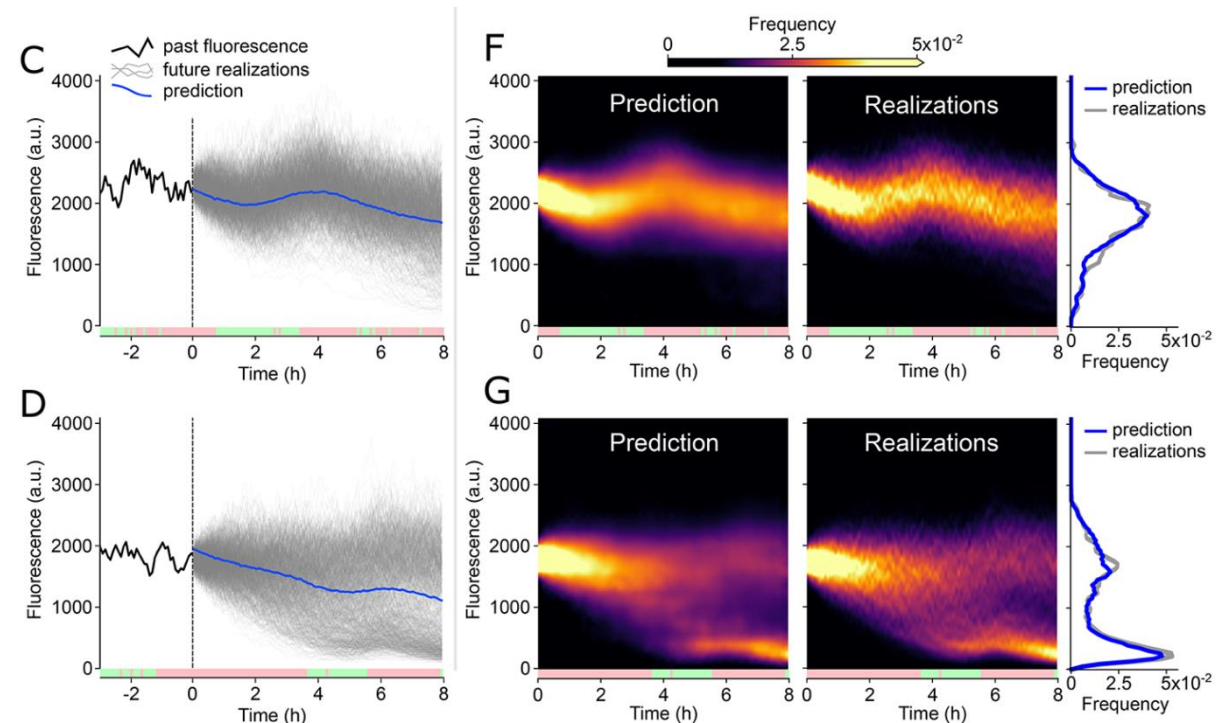
KEYWORDS: time-series prediction, deep learning models, noise, single cell, gene expression dynamics, bistability



Klumpe et al. ACS Synth. Biol. 12, 2367–2381 (2023)

Achieved

- Used LSTM-MLP neural networks to predict future single-cell gene expression trajectories
- Purely computational study, no real experimental data
- Systematically tested factors affecting prediction quality: training set size, amount of past data, circuit complexity (cascade)
- For bistable systems, replaced single-trajectory prediction with an LSTM-CNN that predicts the full distribution of possible future states



Background

Previous publications on deep cell control from Dunlop Lab



pubs.acs.org/synthbio

Research Article

Deep Neural Networks for Predicting Single-Cell Responses and Probability Landscapes

Published as part of the ACS Synthetic Biology virtual special issue “AI for Synthetic Biology”.

Heidi E. Klumpe,^{||} Jean-Baptiste Lugagne,^{*,||} Ahmad S. Khalil, and Mary J. Dunlop^{*}

Cite This: *ACS Synth. Biol.* 2023, 12, 2367–2381

Read Online

ACCESS |

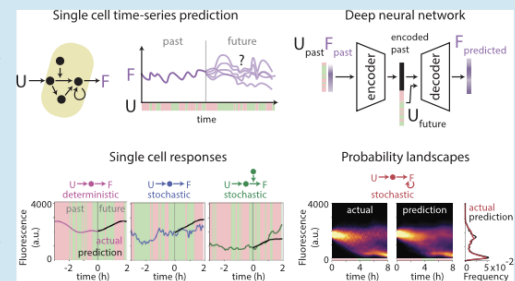
Metrics & More

Article Recommendations

Supporting Information

ABSTRACT: Engineering biology relies on the accurate prediction of cell responses. However, making these predictions is challenging for a variety of reasons, including the stochasticity of biochemical reactions, variability between cells, and incomplete information about underlying biological processes. Machine learning methods, which can model diverse input–output relationships without requiring *a priori* mechanistic knowledge, are an ideal tool for this task. For example, such approaches can be used to predict gene expression dynamics given time-series data of past expression history. To explore this application, we computationally simulated single-cell responses, incorporating different sources of noise and alternative genetic circuit designs. We showed that deep neural networks trained on these simulated data were able to correctly infer the underlying dynamics of a cell response even in the presence of measurement noise and stochasticity in the biochemical reactions. The training set size and the amount of past data provided as inputs both affected prediction quality, with cascaded genetic circuits that introduce delays requiring more past data. We also tested prediction performance on a bistable auto-activation circuit, finding that our initial method for predicting a single trajectory was fundamentally ill-suited for multimodal dynamics. To address this, we updated the network architecture to predict the entire distribution of future states, showing it could accurately predict bimodal expression distributions. Overall, these methods can be readily applied to the diverse prediction tasks necessary to predict and control a variety of biological circuits, a key aspect of many synthetic biology applications.

KEYWORDS: time-series prediction, deep learning models, noise, single cell, gene expression dynamics, bistability



Klumpe et al. *ACS Synth. Biol.* **12**, 2367–2381 (2023)

nature communications



Article

<https://doi.org/10.1038/s41467-024-46361-1>

Deep model predictive control of gene expression in thousands of single cells

Received: 9 August 2023

Accepted: 26 February 2024

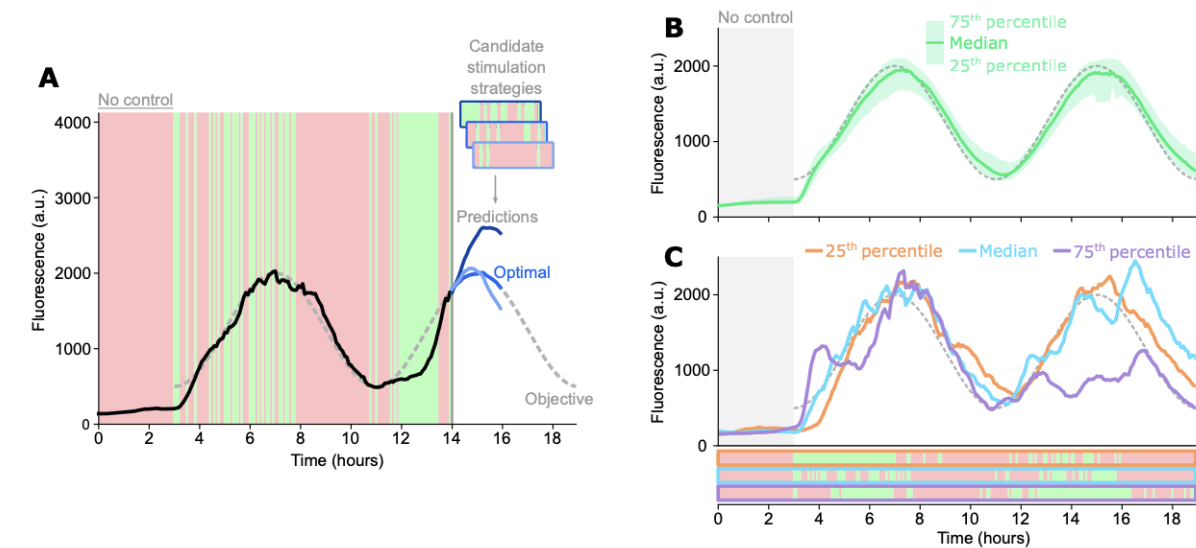
Published online: 08 March 2024

Check for updates

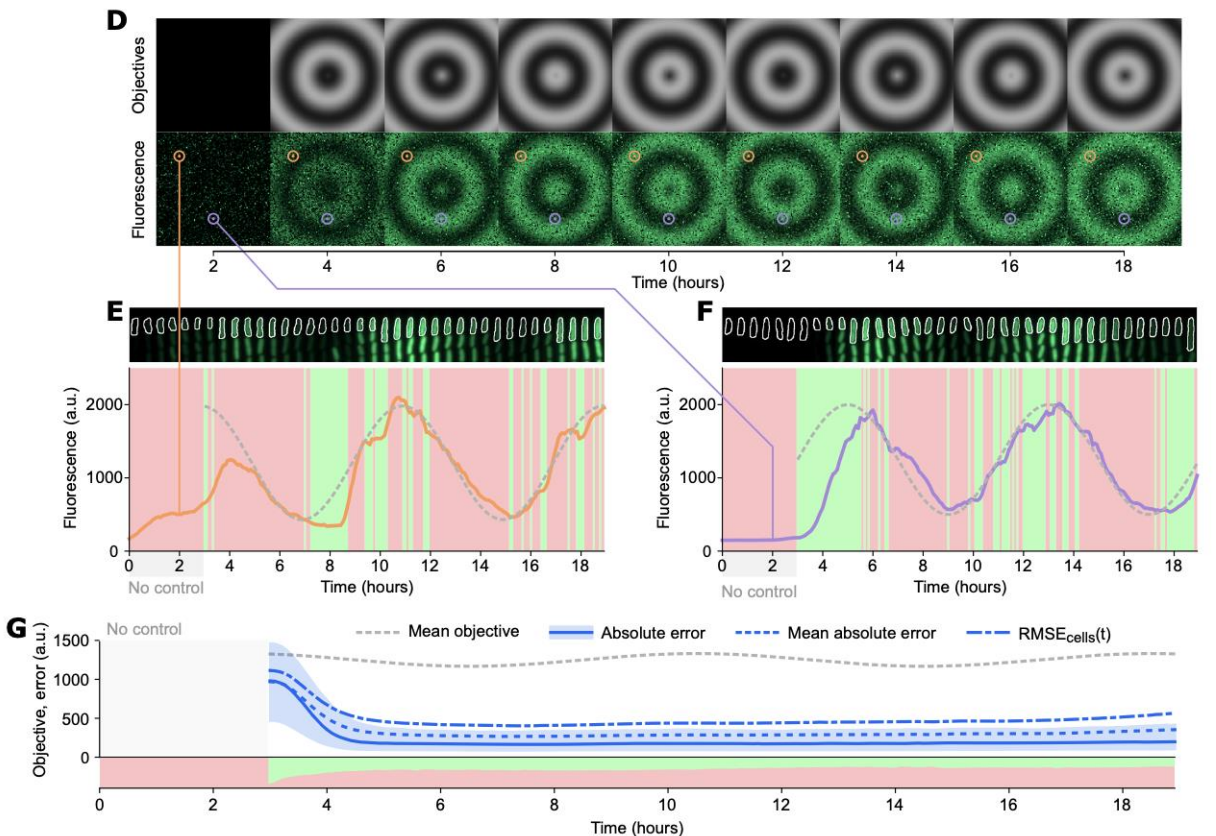
Jean-Baptiste Lugagne^{1,2}✉, Caroline M. Blassick^{1,2} & Mary J. Dunlop^{1,2}✉

Gene expression is inherently dynamic, due to complex regulation and stochastic biochemical events. However, the effects of these dynamics on cell phenotypes can be difficult to determine. Researchers have historically been limited to passive observations of natural dynamics, which can preclude studies of elusive and noisy cellular events where large amounts of data are required to reveal statistically significant effects. Here, using recent advances in the fields of machine learning and control theory, we train a deep neural network to accurately predict the response of an optogenetic system in *Escherichia coli* cells. We then use the network in a deep model predictive control framework to impose arbitrary and cell-specific gene expression dynamics on thousands of single cells in real time, applying the framework to generate complex time-varying patterns. We also showcase the framework's ability to link expression patterns to dynamic functional outcomes by controlling expression of the *tetA* antibiotic resistance gene. This study highlights how deep learning-enabled feedback control can be used to tailor distributions of gene expression dynamics with high accuracy and throughput without expert knowledge of the biological system.

Lugagne, Blassick & Dunlop. *Nat. Commun.* **15**, 2148 (2024)



Every 5 minutes, the model takes each cell's past fluorescence and light history, predicts its response to thousands of candidate light stimulation strategies, and applies the one predicted to best match the target objective. Then repeats.



Background

Previous publications on deep cell control from Dunlop Lab

nature communications



Article

<https://doi.org/10.1038/s41467-024-46361-1>

Deep model predictive control of gene expression in thousands of single cells

Received: 9 August 2023

Accepted: 26 February 2024

Published online: 08 March 2024

Check for updates

Jean-Baptiste Lugagne ^{1,2}✉, Caroline M. Blassick ^{1,2} & Mary J. Dunlop ^{1,2}✉

Gene expression is inherently dynamic, due to complex regulation and stochastic biochemical events. However, the effects of these dynamics on cell phenotypes can be difficult to determine. Researchers have historically been limited to passive observations of natural dynamics, which can preclude studies of elusive and noisy cellular events where large amounts of data are required to reveal statistically significant effects. Here, using recent advances in the fields of machine learning and control theory, we train a deep neural network to accurately predict the response of an optogenetic system in *Escherichia coli* cells. We then use the network in a deep model predictive control framework to impose arbitrary and cell-specific gene expression dynamics on thousands of single cells in real time, applying the framework to generate complex time-varying patterns. We also showcase the framework's ability to link expression patterns to dynamic functional outcomes by controlling expression of the *tetA* antibiotic resistance gene. This study highlights how deep learning-enabled feedback control can be used to tailor distributions of gene expression dynamics with high accuracy and throughput without expert knowledge of the biological system.

Lugagne, Blassick & Dunlop. *Nat. Commun.* **15**, 2148 (2024)

Achieved

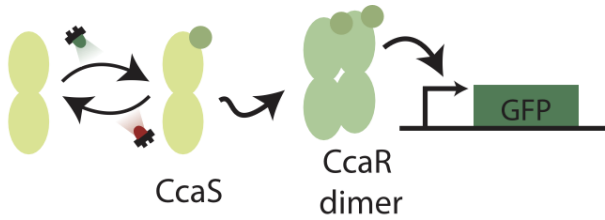
- Trained a DNN to predict single-cell gene expression responses, outperforming traditional ODE + Kalman filter approaches
- Integrated the prediction model into an MPC framework for real-time closed-loop control
- Simultaneously controlled ~10,000 single cells, improving throughput by 2 orders of magnitude over previous work
- Assigned each cell an independent, arbitrary target trajectory with high tracking accuracy

Limitation

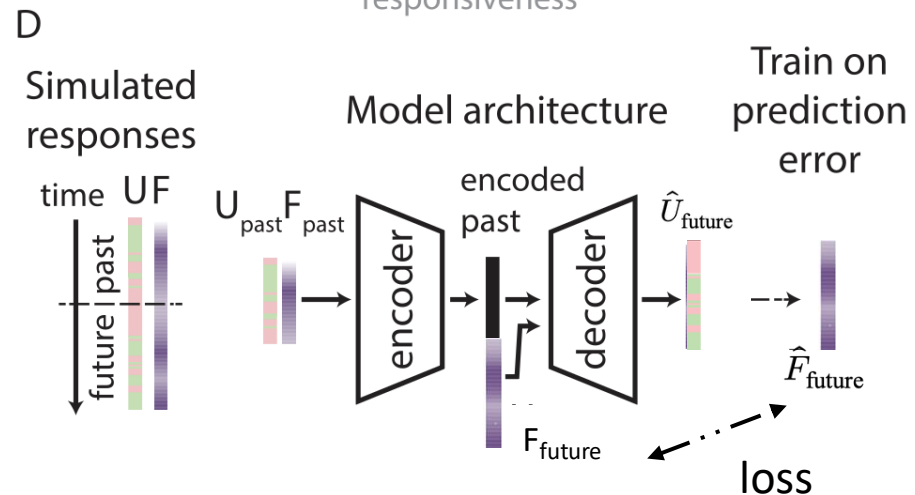
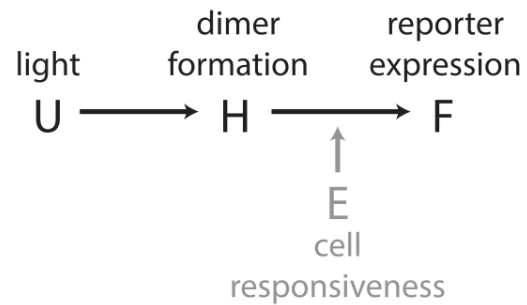
Not end-to-end: the model predicts responses but requires a separate optimizer to search over candidate strategies, making control computationally expensive

Method

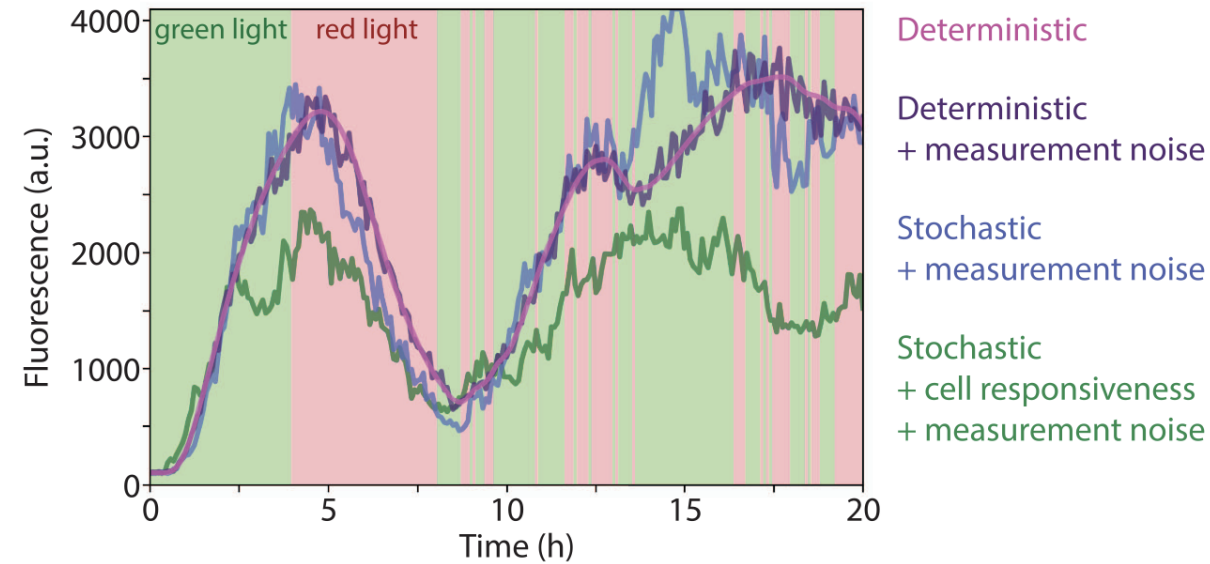
A Example inducible system: CcaSR



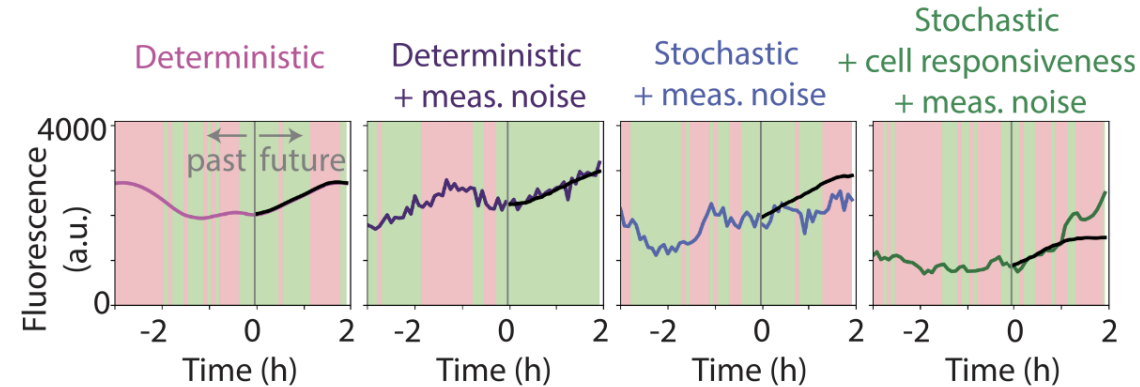
B Circuit architecture



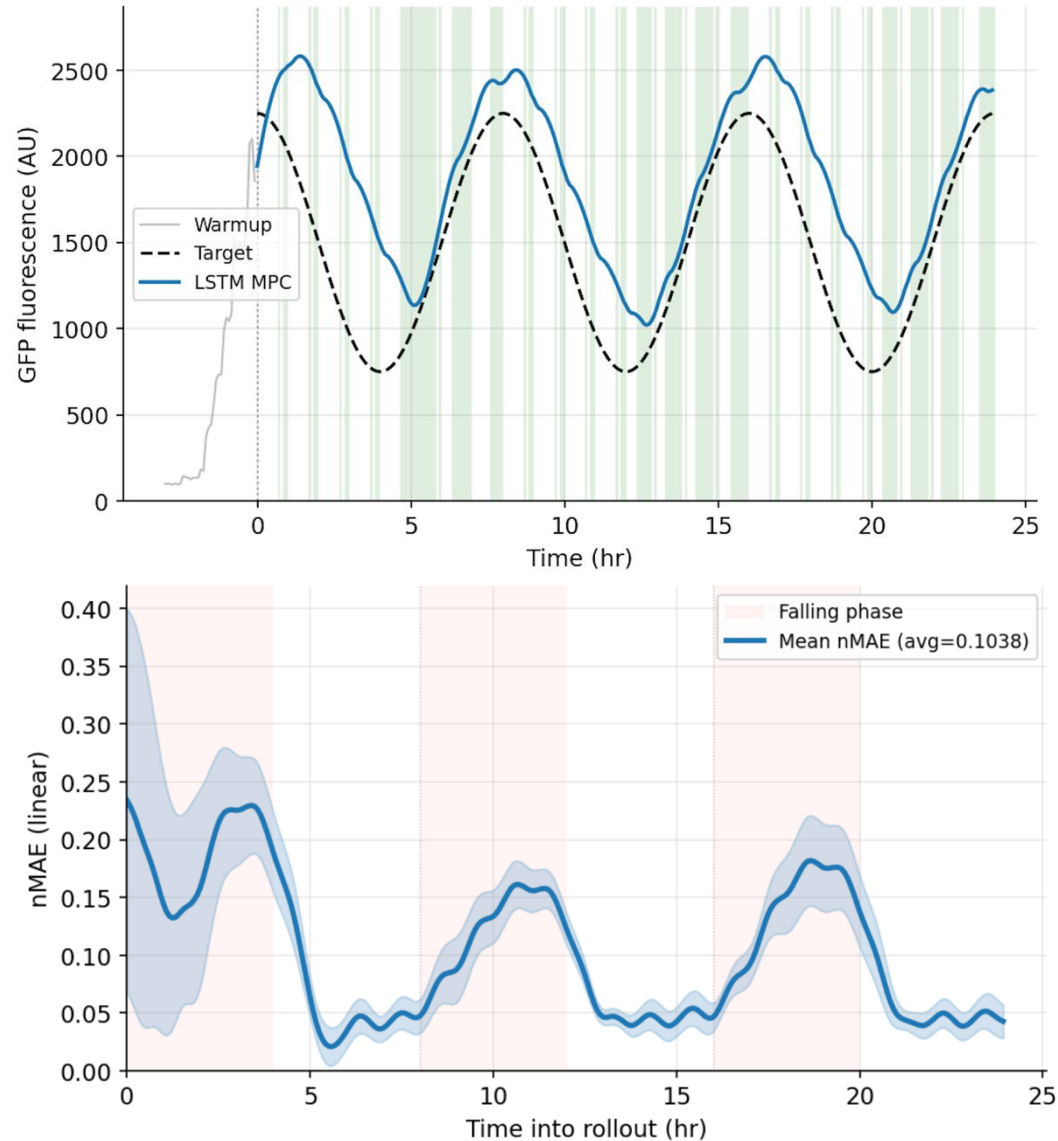
C Sample simulations



E Median prediction examples



Result



Setup

- Train on 100 cells, simulate by (Stochastic + E + measurement noise), use future simulated cell response as target
- Predict on 10 cell, for each cell we simulate cell response of same predicted control signal by (Stochastic + E + measurement noise) for 10 times to make nMAE more stable, use a 8h period cosine as target.

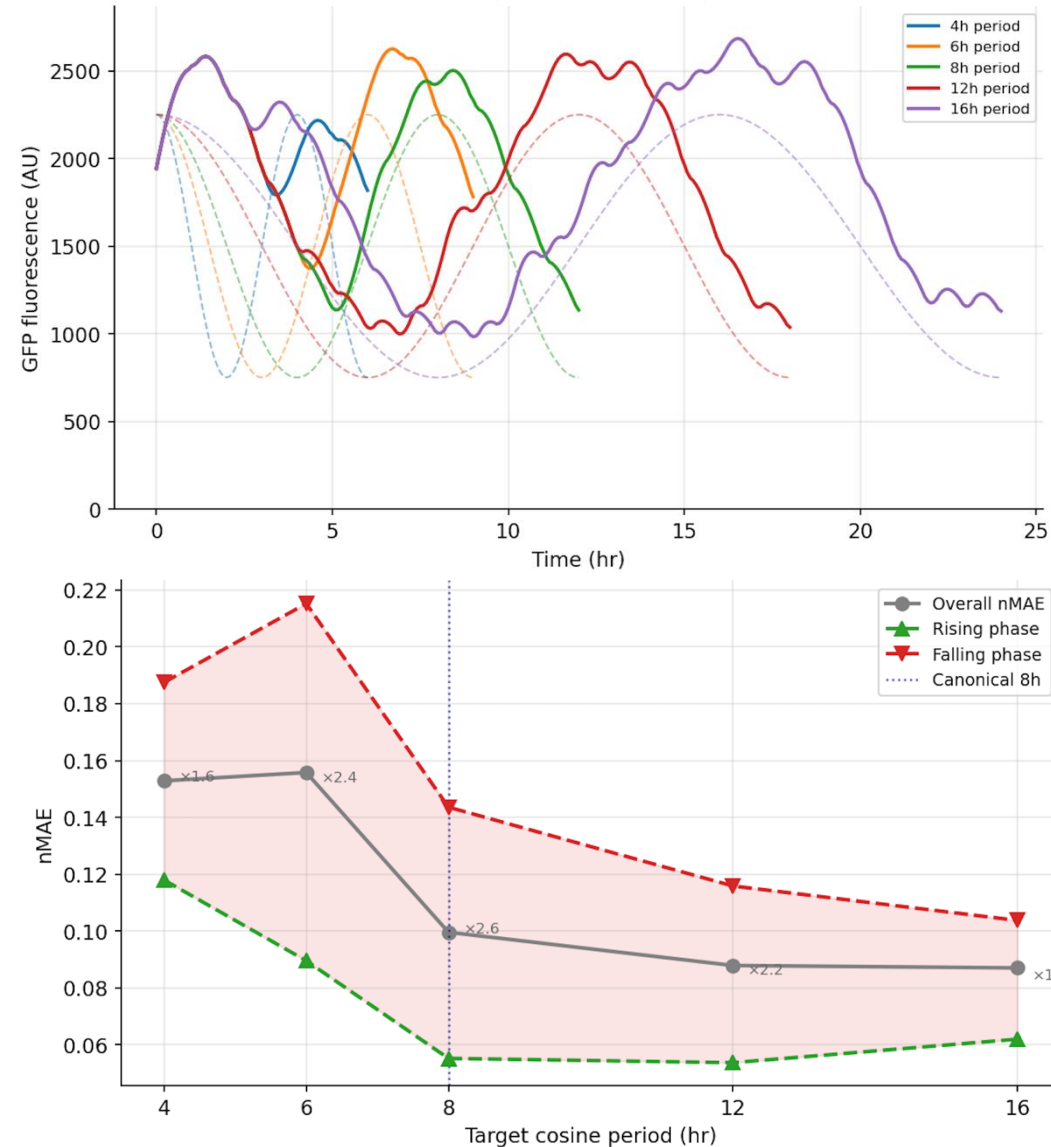
Figures

- *Top*: example single-cell prediction.
- *Bottom*: mean nMAE over rollout time, averaged across 10 cells.

Findings

1. Initial error is high due to the gap between the cell's starting GFP level and the target
2. Model tracks rising periods more accurately than falling periods, as GFP reduction relies on passive dilution rather than active degradation
3. nMAE variance across cells increases over time, reflecting accumulating stochasticity

Result: evaluation on error relate to cosine target period



Setup

Same as previous, except use cosine with different period as target.

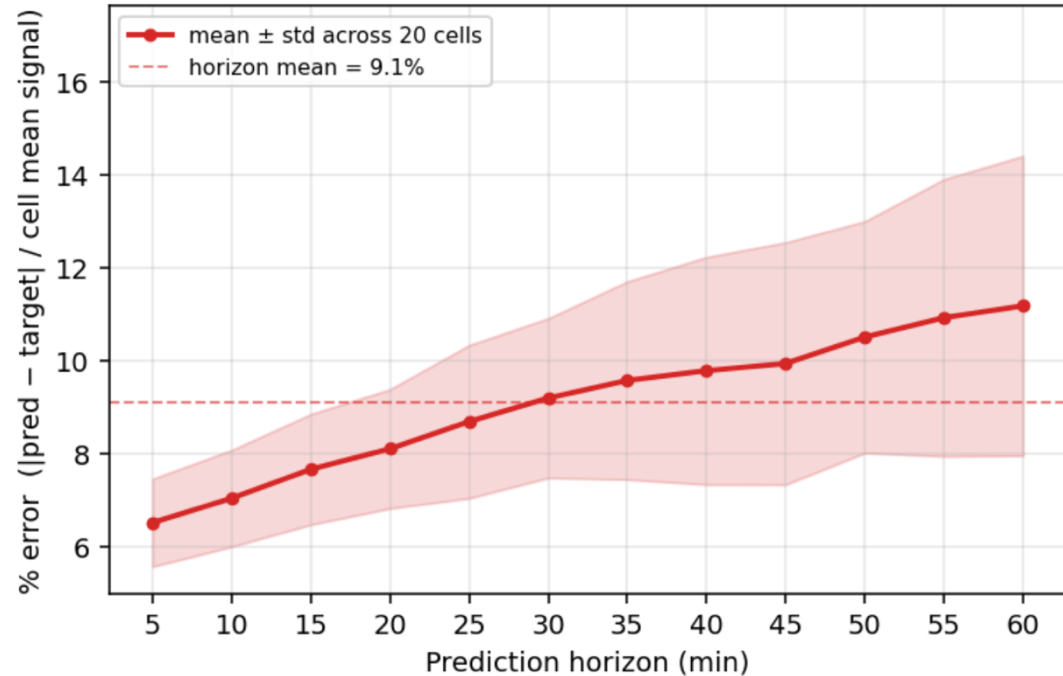
Figure

- Top*: Example rollout of a single cell tracking cosine targets across five periods.
solid lines show simulated cell response, dashed lines show target
- Bottom*: Mean nMAE across 10 cells, split by rising and falling phase, as a function of target period.
annotated ratios show falling/rising error at each period

Findings

- Overall nMAE decreases as period increases, i.e., slower targets are easier to track, as the cell has more time to respond
- Falling phase consistently outperforms rising across all periods (x1.6–x2.6), i.e., asymmetry between active transcription and passive dilution-driven decay

Result: evaluation on error relate to time



Setup

Same as previous, except the target is no longer a predefined cosine; we simulate 20 cell response and use its raw trajectory as the tracking target.

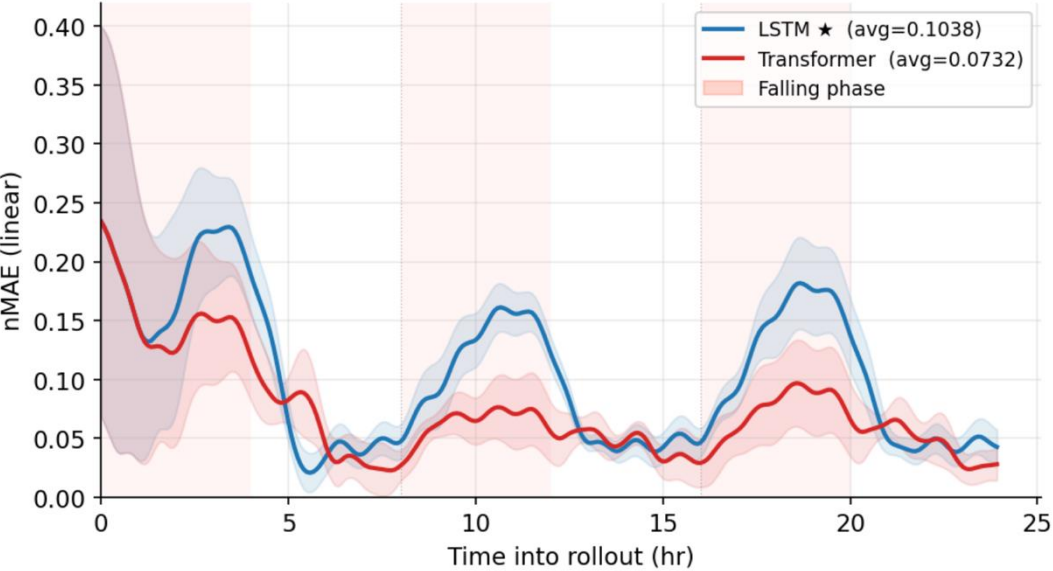
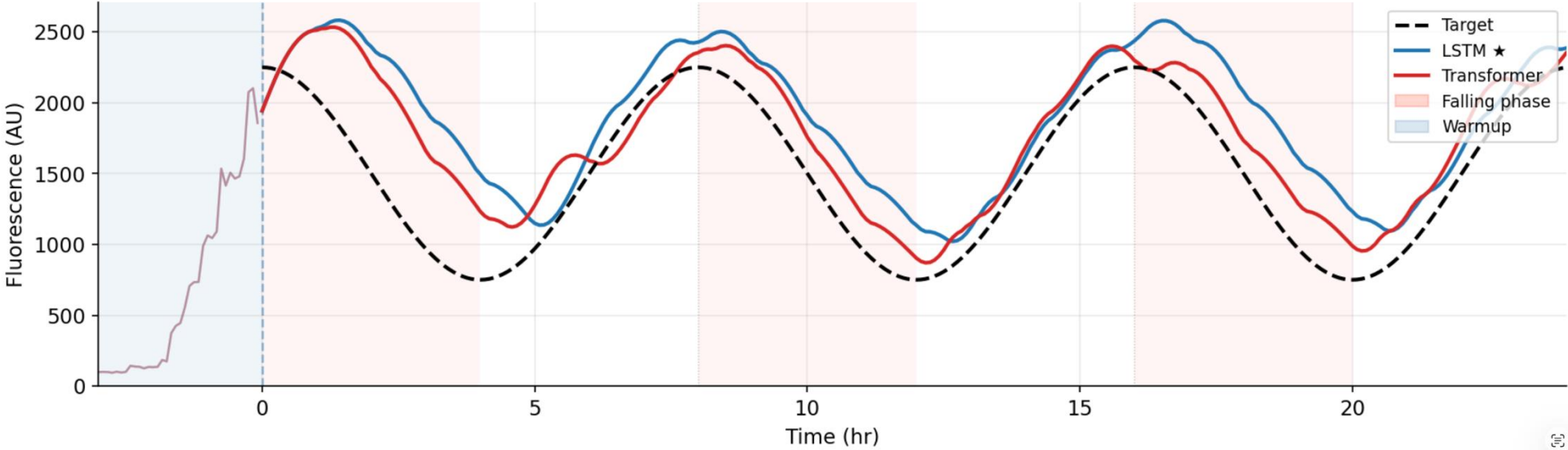
Figures

nMAE as a function of rollout horizon, averaged across 20 cells.

Findings

Error increases monotonically with prediction horizon, reflecting accumulating stochasticity over time.

Result: ablation on model architectures



Setup

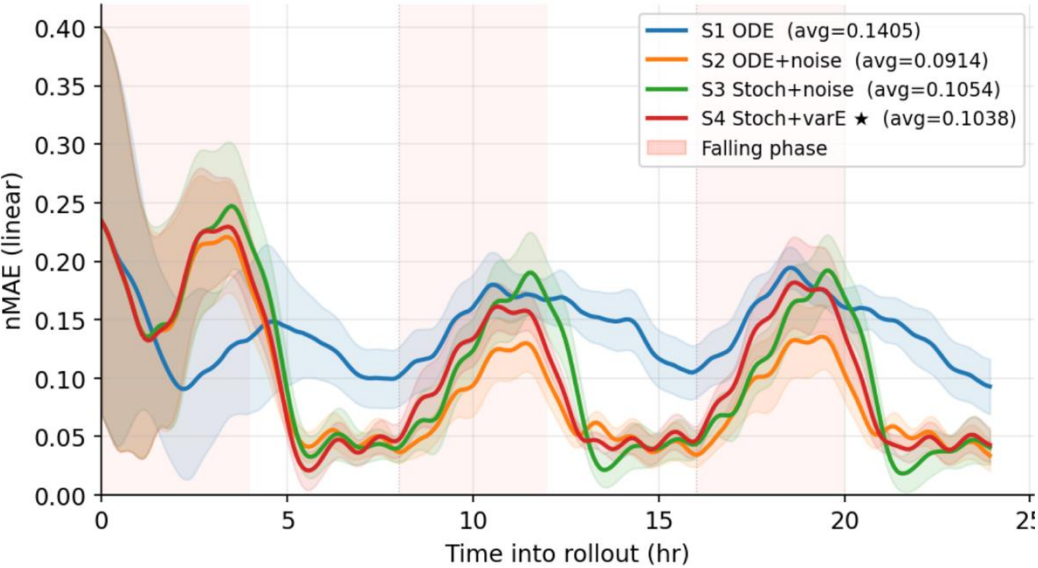
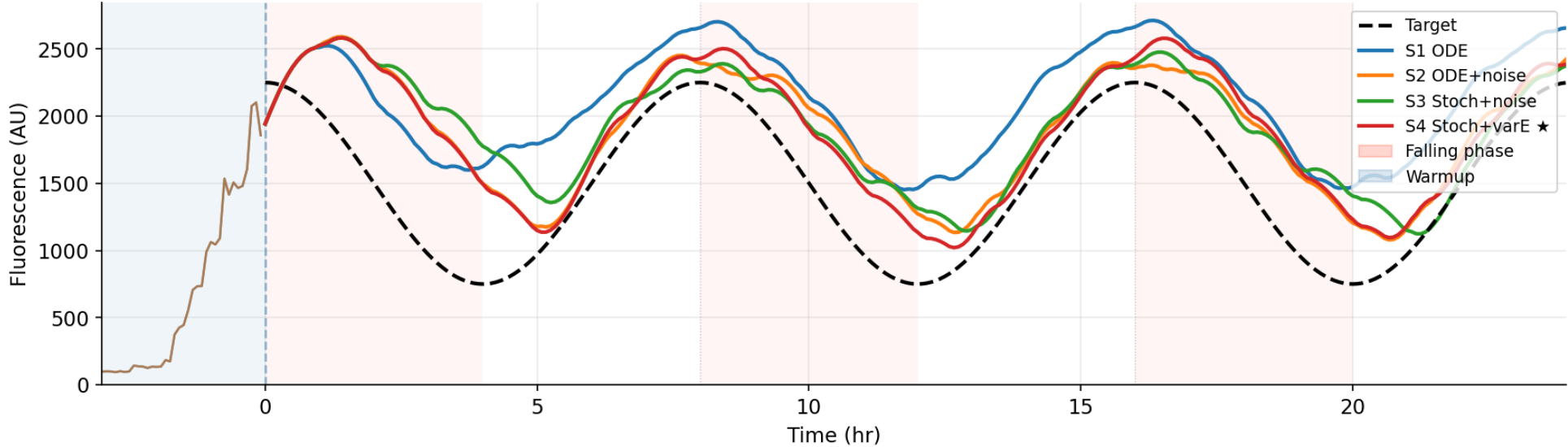
Same setup as first result. Two model architectures are compared: LSTM encoder and Transformer encoder, with all other components held constant.

Figure

- *Top*: Example single-cell rollout over ~24h.
- *Bottom*: nMAE over rollout time.

colored by model architectures

Result: ablation on data simulation method



Setup

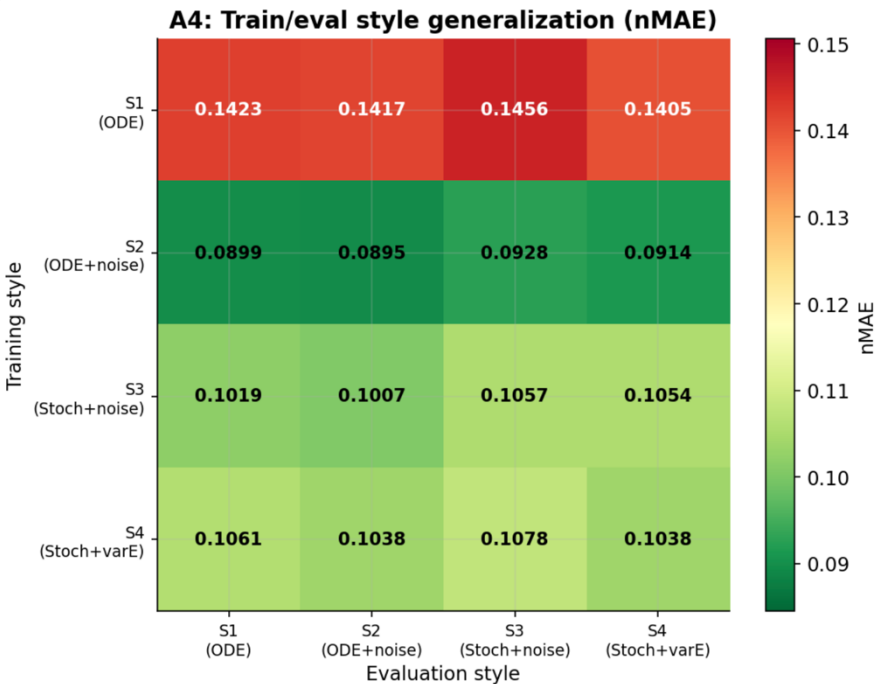
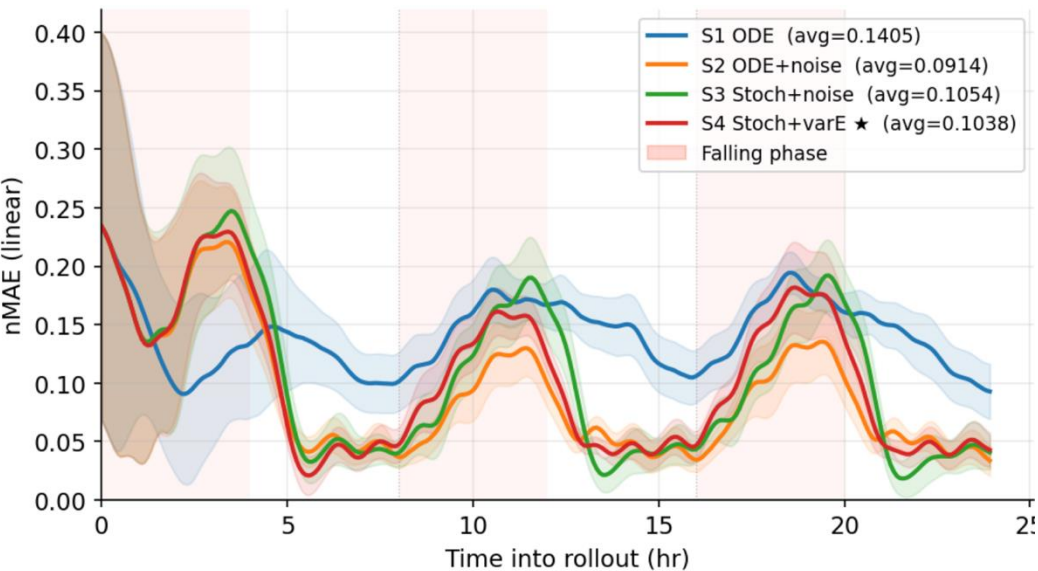
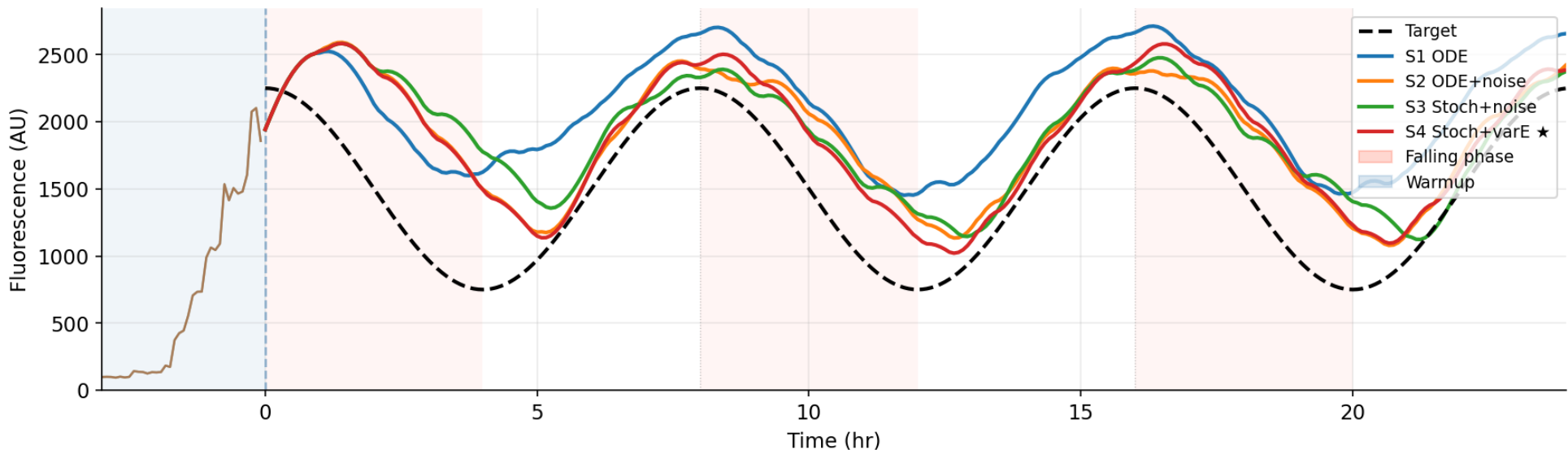
Same setup as first result, except we use different method to simulate the training data.

Figure

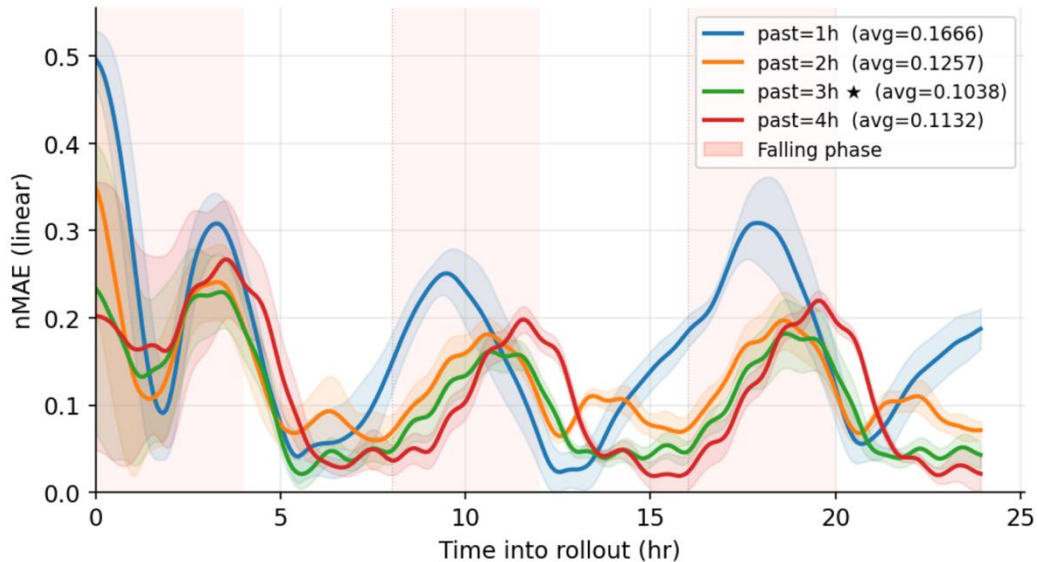
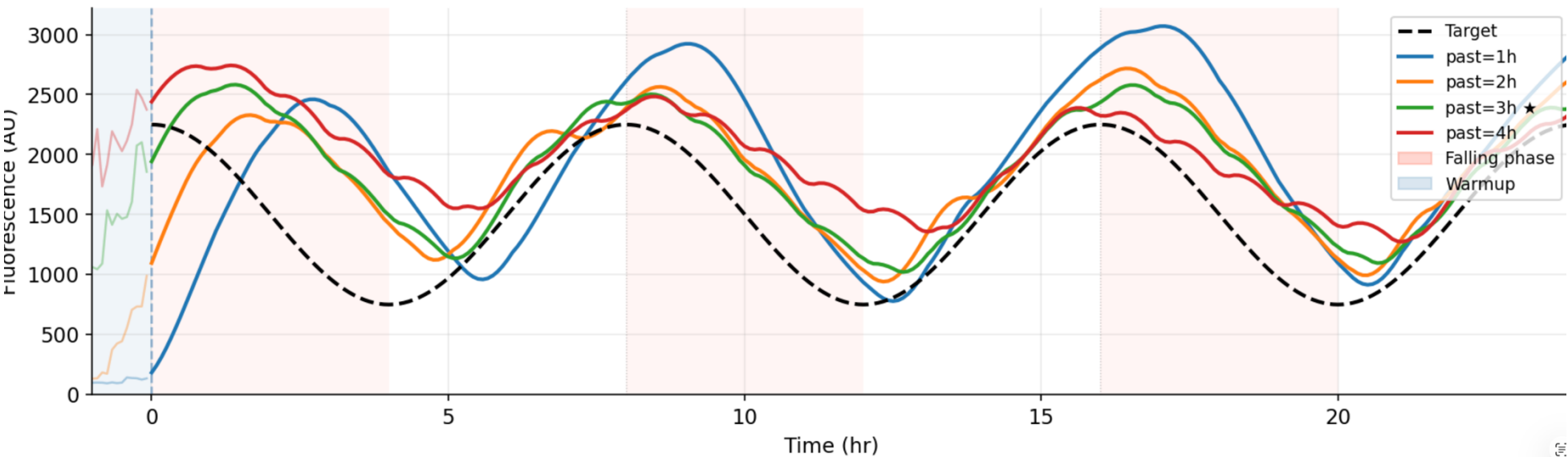
- *Top*: Example single-cell rollout over ~24h.
- *Bottom*: nMAE over rollout time.

colored by simulation method

Result: ablation on data simulation method



Result: ablation on input length



Setup

Same setup as first result, except we use different input length.

Figure

- *Top*: Example single-cell rollout over ~24h.
- *Bottom*: nMAE over rollout time.
colored by input length